

Exercise 4, Part I:

Part(i) Volume of Ellipsoid:

```
a=4;b=2;c=1;
V_analy = (4*pi*a*b*c)/3; % analytical volume
lowerlim = {-a, @ylow, @zlow};
upperlim = {a,@yup,@zup};
I = gsimp(@ellip,lowerlim,upperlim,200);
perror = 100*abs((I-V_analy)/V_analy);
fprintf(['Analytical value = %f \n' ...
        'Numerical value using simpsons %f \n' ...
        'Percent error %f \n'],V_analy,I,perror)
```

```
Anlytical value = 33.510322
Numerical value using simpsons 33.505421
Percent error 0.014625
```

The result obtained by the numerical approximation, 33.505 is close to the analytical value of 33.51 with a percent error of 0.0146%. This means that the numerical value is less than 0.015% away from the analytical value.

```
function [v] = ellip(X)
x=X(1);y=X(2);z=X(3);
v = 1;
end

function [v]=zlow(X) % z lower limit
x=X(1);y=X(2);
a=4;b=2;c=1;
v = -c*sqrt(1-((x/a)^2)-((y/b)^2));
end

function [v]=zup(X)
x=X(1);y=X(2);
a=4;b=2;c=1;
v = c*sqrt(1-((x/a)^2)-((y/b)^2));
end

function [v] = ylow(X)
x=X(1);
a=4;b=2;
v = -b*sqrt(1-((x/a)^2));
end

function [v] = yup(X)
x=X(1);
a=4;b=2;
v = b*sqrt(1-((x/a)^2));
end

% multi-dimensional simpson main calling function
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```

% goes from outer to inner (i.e., x to z).
function [I] = gsimp(f,a,b,n)
m=length(a); x=zeros(1,m); k=1;
% convert limits to cell array if numbers are given
if (isnumeric(a))
    a=num2cell(a);
end
if (isnumeric(b))
    b=num2cell(b);
end
% call recursive integration function
I=ksimp(f,x,a,b,n,k);
end

% recursive simpson function
function [I] = ksimp(f,x,a,b,n,k)
I=0; m=length(a);
if isnumeric(a{k})
    ak=a{k};
else
    ak=feval(a{k},x);
end
if isnumeric(b{k})
    bk=b{k};
else
    bk=feval(b{k},x);
end
h =(bk-ak)/n;
for i=0:n
    x(k)=ak+i*h;
    if (k == m)
        if (i == 0 || i == n)
            I=I+feval(f,x);
        elseif (mod(i,2) == 1)
            I=I+4*feval(f,x);
        elseif (mod(i,2) == 0)
            I=I+2*feval(f,x);
        end
    else
        if (i == 0 || i == n)
            I=I+ksimp(f,x,a,b,n,k+1);
        elseif (mod(i,2) == 1)
            I=I+4*ksimp(f,x,a,b,n,k+1);
        elseif (mod(i,2) == 0)
            I=I+2*ksimp(f,x,a,b,n,k+1);
        end
    end
end
I=I*h/3;
end

```